GFE0007 Data Analytics Made Easy: For Beginner

Analytics on Life Expectancy

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1. About Dataset

1.1 Context

Although there have been lot of studies undertaken in the past on factors affecting life expectancy considering demographic variables, income composition and mortality rates. It was found that affect of immunization and human development index was not taken into account in the past. Also, some of the past research was done considering multiple linear regression based on data set of one year for all the countries. Hence, this gives motivation to resolve both the factors stated previously by formulating a regression model based on mixed effects model and multiple linear regression while considering data from a period of 2000 to 2015 for all the countries. Important immunization like Hepatitis B, Polio and Diphtheria will also be considered. In a nutshell, this study will focus on immunization factors, mortality factors, economic factors, social factors and other health related factors as well. Since the observations this dataset are based on different countries, it will be easier for a country to determine the predicting factor which is contributing to lower value of life expectancy. This will help in suggesting a country which area should be given importance in order to efficiently improve the life expectancy of its population.

1.2 Content

The project relies on accuracy of data. The Global Health Observatory (GHO) data repository under World Health Organization (WHO) keeps track of the health status as well as many other related factors for all countries The data-sets are made available to public for the purpose of health data analysis. The data-set related to life expectancy, health factors for 193 countries has been collected from the same WHO data repository website and its corresponding economic data was collected from United Nation website. Among all categories of health-related factors only those critical factors were chosen which are more representative. It has been observed that in the past 15 years, there has been a huge development in health sector resulting in improvement of human mortality rates especially in the developing nations in comparison to the past 30 years. Therefore, this project has been considered data from year 2000-2015 for 193 countries for further

analysis. The individual data files have been merged together into a single data-set. On initial visual inspection of the data showed some missing values. As the data-sets were from WHO, no evident errors were found. Missing data was handled in R software by using Missmap command. The result indicated that most of the missing data was for population, Hepatitis B and GDP. The missing data were from less known countries like Vanuatu, Tonga, Togo, Cabo Verde etc. Finding all data for these countries was difficult and hence, it was decided that these countries were excluded from the final model data-set. The final merged file (final dataset) consists of 22 Columns and 2938 rows which meant 20 predicting variables. All predicting variables was then divided into several broad categories: Immunization related factors, Mortality factors, Economical factors and Social factors.

2. Data Requirement Gathering

2.1 Objective

The project will discover on what are the main factor of life expectancy and how strong is the relationship on them. The question comes up for this project is, "**what is the relationship between income composition of resources and life expectancy?**". The hypothesis for this project is, "**when income composition of resources increases, then life expectancy will also increase**".

2.2 Tools

The tool used for this project is R. It is a language and environment for statistical computing and graphics. It provides a wide variety of statistical such as linear and nonlinear modelling, classical statistical tests, time-series analysis, classification, clustering, etc., and graphical techniques, and it is highly extensible. One of R's strengths is the ease with which well-designed publication-quality plots can be produced, including mathematical symbols and formulae where needed. Integrated Development Environment (IDE) that will be using for coding in R language is RStudio.

3. Data Collection

3.1 Selecting Dataset

The dataset will be using is Life Expectancy Dataset published by Kumar Rajarshi on Kaggle website. The reason this dataset is chosen because this dataset has many continuous variables so that it would be easy to do analysis instead of converting every categorical variable into numeric. Next is because this dataset has thousands of rows and many columns which is good to explore more features on this dataset and correlation between them, and get more detailed results for this project.

3.2 Explore Dataset

After downloading the dataset, import it into RStudio by running the code in Figure 1. Basically, RStudio will let the user to select which Excel file and it read the file, then later store into a variable as a dataset.

```
life_exp <- read.csv(file.choose())
```

Figure 1: Import dataset code

After that, will be using *glimpse( )* and *descr( )* to understand dataset better in depth. *glimpse( )* function is a little like *str( )* applied to a data frame but it tries to show you as much data as possible. *descr( )* is function from *summarytools* library that produces descriptive (univariate) statistics with common central tendency statistics and measures of dispersion. A major advantage of this function is that it accepts single vectors as well as data frames. If a data frame is provided, all non-numerical columns are ignored so you do not have to remove them yourself before running the function.

```
> glimpse(life_exp)
Rows: 2,938
Columns: 22
$ Country                        <chr> "Afghanistan", "Afghanistan", "Afghanistan", "Afghanistan", "Afghanistan", "Afghanistan", "Afghanistan", "Afghanistan", "Afghan…
$ Year                           <int> 2015, 2014, 2013, 2012, 2011, 2010, 2009, 2008, 2007, 2006, 2005, 2004, 2003, 2002, 2001, 2000, 2015, 2014, 2013, 2012, 2011, 2…
$ Status                         <chr> "Developing", "Developing", "Developing", "Developing", "Developing", "Developing", "Developing", "Developing", "Developing", "…
$ Life.expectancy                <dbl> 65.0, 59.9, 59.9, 59.5, 59.2, 58.8, 58.6, 58.1, 57.5, 57.3, 57.3, 57.0, 56.7, 56.2, 55.3, 54.8, 77.8, 77.5, 77.2, 76.9, 76.6, 7…
$ Adult.Mortality                <int> 263, 271, 268, 272, 275, 279, 281, 287, 295, 295, 291, 293, 295, 3, 316, 321, 74, 8, 84, 86, 88, 91, 91, 1, 9, 99, 15, 17, 18, …
$ infant.deaths                  <int> 62, 64, 66, 69, 71, 74, 77, 80, 82, 84, 85, 87, 87, 88, 88, 88, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 21, 21, 21, 21,…
$ Alcohol                        <dbl> 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.03, 0.02, 0.03, 0.02, 0.02, 0.01, 0.01, 0.01, 0.01, 4.60, 4.51, 4.76, 5.14, 5.37, 5…
$ percentage.expenditure         <dbl> 71.279624, 73.523582, 73.219243, 78.184215, 7.097109, 79.679367, 56.762217, 25.873925, 10.910156, 17.171518, 1.388648, 15.29606…
$ Hepatitis.B                    <int> 65, 62, 64, 67, 68, 66, 63, 64, 63, 64, 66, 67, 65, 64, 63, 62, 99, 98, 99, 99, 99, 99, 98, 99, 98, 98, 98, 99, 97, 96, 96, 96,…
$ Measles                        <int> 1154, 492, 430, 2787, 3013, 1989, 2861, 1599, 1141, 1990, 1296, 466, 798, 2486, 8762, 6532, 0, 0, 0, 9, 28, 10, 0, 0, 22, 68, 6…
$ BMI                            <dbl> 19.1, 18.6, 18.1, 17.6, 17.2, 16.7, 16.2, 15.7, 15.2, 14.7, 14.2, 13.8, 13.4, 13.0, 12.6, 12.2, 58.0, 57.2, 56.5, 55.8, 55.1, 5…
$ under.five.deaths              <int> 83, 86, 89, 93, 97, 102, 106, 110, 113, 116, 118, 120, 122, 122, 122, 122, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 24, …
$ Polio                          <int> 6, 58, 62, 67, 68, 66, 63, 64, 63, 58, 58, 5, 41, 36, 35, 24, 99, 98, 99, 99, 99, 99, 98, 99, 99, 97, 97, 98, 97, 98, 97, 97, 9…
$ Total.expenditure              <dbl> 8.16, 8.18, 8.13, 8.52, 7.87, 9.20, 9.42, 8.33, 6.73, 7.43, 8.70, 8.79, 8.82, 7.76, 7.80, 8.20, 6.00, 5.88, 5.66, 5.59, 5.71, 5…
$ Diphtheria                     <int> 65, 62, 64, 67, 68, 66, 63, 64, 63, 58, 58, 5, 41, 36, 33, 24, 99, 98, 99, 99, 99, 99, 98, 99, 98, 97, 98, 97, 97, 98, 97, 97, …
$ HIV.AIDS                       <dbl> 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.1, 0.…
$ GDP                            <dbl> 584.25921, 612.69651, 631.74498, 669.95900, 63.53723, 553.32894, 445.89330, 373.36112, 369.83580, 272.56377, 25.29413, 219.1413…
$ Population                     <dbl> 33736494, 327582, 31731688, 3696958, 2978599, 2883167, 284331, 2729431, 26616792, 2589345, 257798, 24118979, 2364851, 21979923,…
$ thinness..1.19.years           <dbl> 17.2, 17.5, 17.7, 17.9, 18.2, 18.4, 18.6, 18.8, 19.0, 19.2, 19.3, 19.5, 19.7, 19.9, 2.1, 2.3, 1.2, 1.2, 1.3, 1.3, 1.4, 1.4, 1.5…
$ thinness.5.9.years             <dbl> 17.3, 17.5, 17.7, 18.0, 18.2, 18.4, 18.7, 18.9, 19.1, 19.3, 19.5, 19.7, 19.9, 2.2, 2.4, 2.5, 1.3, 1.3, 1.4, 1.4, 1.5, 1.5, 1.6,…
$ Income.composition.of.resources <dbl> 0.479, 0.476, 0.470, 0.463, 0.454, 0.448, 0.434, 0.433, 0.415, 0.405, 0.396, 0.381, 0.373, 0.341, 0.340, 0.338, 0.762, 0.761, 0…
$ Schooling                      <dbl> 10.1, 10.0, 9.9, 9.8, 9.5, 9.2, 8.9, 8.7, 8.4, 8.1, 7.9, 6.8, 6.5, 6.2, 5.9, 5.5, 14.2, 14.2, 14.2, 14.2, 13.3, 12.5, 12.2, 12.…
```

Figure 2: glimpse( )

```
> descr(life_exp)
Non-numerical variable(s) ignored: Country, Status
Descriptive Statistics
life_exp
N: 2938
```

	Adult.Mortality	Alcohol	BMI	Diphtheria	GDP	Hepatitis.B	HIV.AIDS	Income.composition.of.resources	infant.deaths	Life.expectancy
Mean	164.80	4.60	38.32	82.32	7483.16	80.94	1.74	0.63	30.30	69.22
Std.Dev	124.29	4.05	20.04	23.72	14270.17	25.07	5.08	0.21	117.93	9.52
Min	1.00	0.01	1.00	2.00	1.68	1.00	0.10	0.00	0.00	36.30
Q1	74.00	0.88	19.30	78.00	463.85	77.00	0.10	0.49	0.00	63.10
Median	144.00	3.76	43.50	93.00	1766.95	92.00	0.10	0.68	3.00	72.10
Q3	228.00	7.70	56.20	97.00	5918.20	97.00	0.80	0.78	22.00	75.70
Max	723.00	17.87	87.30	99.00	119172.74	99.00	50.60	0.95	1800.00	89.00
MAD	112.68	4.81	24.17	8.90	2360.98	8.90	0.00	0.19	4.45	8.60
IQR	154.00	6.82	36.90	19.00	5446.87	20.00	0.70	0.29	22.00	12.60
CV	0.75	0.88	0.52	0.29	1.91	0.31	2.91	0.34	3.89	0.14
Skewness	1.17	0.59	-0.22	-2.07	3.20	-1.93	5.39	-1.14	9.78	-0.64
SE.Skewness	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Kurtosis	1.74	-0.81	-1.29	3.55	12.29	2.76	34.80	1.38	115.76	-0.24
N.Valid	2928.00	2744.00	2904.00	2919.00	2490.00	2385.00	2938.00	2771.00	2938.00	2928.00
Pct.Valid	99.66	93.40	98.84	99.35	84.75	81.18	100.00	94.32	100.00	99.66

	Measles	percentage.expenditure	Polio	Population	Schooling	thinness..1.19.years	thinness.5.9.years	Total.expenditure	under.five.deaths	Year
Mean	2419.59	738.25	82.55	12753375.12	11.99	4.84	4.87	5.94	42.04	2007.52
Std.Dev	11467.27	1987.91	23.43	61012096.51	3.36	4.42	4.51	2.50	160.45	4.61
Min	0.00	0.00	3.00	34.00	0.00	0.10	0.10	0.37	0.00	2000.00
Q1	0.00	4.67	78.00	195553.00	10.10	1.60	1.50	4.26	0.00	2004.00
Median	17.00	64.91	93.00	1386542.00	12.30	3.30	3.30	5.76	4.00	2008.00
Q3	361.00	441.84	97.00	7423289.00	14.30	7.20	7.20	7.50	28.00	2012.00
Max	212183.00	19479.91	99.00	1293859294.00	20.70	27.70	28.60	17.60	2500.00	2015.00
MAD	25.20	96.24	8.90	2012347.06	3.11	3.41	3.41	2.36	5.93	5.93
IQR	360.25	436.85	19.00	7224565.75	4.20	5.60	5.70	3.23	28.00	8.00
CV	4.74	2.69	0.28	4.78	0.28	0.91	0.93	0.42	3.82	0.00
Skewness	9.43	4.65	-2.10	15.90	-0.60	1.71	1.78	0.62	9.49	-0.01
SE.Skewness	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Kurtosis	114.58	26.51	3.76	297.09	0.88	3.96	4.34	1.15	109.49	-1.21
N.Valid	2938.00	2938.00	2919.00	2286.00	2775.00	2904.00	2904.00	2712.00	2938.00	2938.00
Pct.Valid	100.00	100.00	99.35	77.81	94.45	98.84	98.84	92.31	100.00	100.00

Figure 3: descr( )

4. Data Cleaning

4.1 Replacing N/A with Mean Group by Country

The first step is to count how many null values are there in all column. This can be achieved by using *colSums( )* and is.na( ) function. *colSums()* computes the sums of matrix or array columns while *is.na( )* deal with missing values in the dataset or data frame.

```
> colSums(is.na(life_exp))
                  Country                     Year                   Status             Life.expectancy             Adult.Mortality
                        0                        0                        0                          10                          10
             infant.deaths                  Alcohol     percentage.expenditure                        0                 Hepatitis.B                     Measles
                        0                      194                        0                      553                           0
                      BMI        under.five.deaths                    Polio        Total.expenditure                   Diphtheria
                       34                        0                       19                      226                          19
                 HIV.AIDS                      GDP               Population       thinness..1.19.years         thinness.5.9.years
                        0                      448                      652                       34                          34
Income.composition.of.resources          Schooling
                      167                      163
```

Figure 4: Count null values

After that, null values will be replaced by the mean respective by their column group by country. For example, the mean for Alcohol column is 4.60. If China has a row that has missing value in Alcohol, instead of replacing with 4.60, it will be replaced with the mean of Alcohol in China only. This operation will be using *group_by( ), mutate_at( ),*

vars( ), and *replace_na( ).* What this code does is to group the dataset by Country, add new columns that are selected and later replace null values with their grouped mean.

```
> le_cleaned <- le_copy %>%
+   group_by(Country) %>%
+   mutate_at(vars(Life.expectancy, Adult.Mortality, Alcohol, Hepatitis.B, BMI,
+                  Polio, Total.expenditure, Diphtheria, GDP, Population,
+                  thinness..1.19.years, thinness.5.9.years,
+                  Income.composition.of.resources, Schooling), ~replace_na(., mean(., na.rm = TRUE)))
```

Figure 5: Replacing N/A with mean group by country

Then, double confirmation is done to check whether the implementation is correct or not. From the figure below, it shows that the replacement value of missing value of Alcohol from Angola with mean values from its country is correct.

```
> # Before Replace N/A
> le_copy$Alcohol[le_copy$Country == 'Angola' & le_copy$Year == 2015]
[1] NA
> # After Replace N/A with mean Group by Country
> le_cleaned$Alcohol[le_cleaned$Country == 'Angola' & le_copy$Year == 2015]
[1] 5.740667
> # Calculate Angola's Alcohol Mean
> Angola_db <- le_copy[le_copy$Country == 'Angola',]
> mean(Angola_db$Alcohol, na.rm = TRUE)
[1] 5.740667
```

Figure 6: Double check

4.2 Replacing N/A with Mean

When counting the number of null values again for columns, it shows that there are few total number left even after replacing with mean. It shows that there may have a country that has null values for the entire column. For this problem, the solution would be replacing the null values with overall mean respective to their column. The code is similar with Figure 5, only remove the group_by(Country) line. Once it's done, the system finally shows that there are no null values left in the dataset.

```
> colSums(is.na(le_cleanedv2))
                     Country                        Year                      Status               Life.expectancy               Adult.Mortality
                           0                           0                           0                             0                             0
                infant.deaths                     Alcohol       percentage.expenditure                  Hepatitis.B                      Measles
                           0                           0                           0                             0                             0
                         BMI             under.five.deaths                       Polio             Total.expenditure                   Diphtheria
                           0                           0                           0                             0                             0
                    HIV.AIDS                         GDP                  Population          thinness..1.19.years           thinness.5.9.years
                           0                           0                           0                             0                             0
Income.composition.of.resources                  Schooling
                           0                           0
```

Figure 7: No N/A value

4.3 Winsorizing

After replacing null values, the next values to be replacing is outliers. One of many ways to spot the outliers are by using boxplot. *ggplot( )* is used to generate the plot while *geom_boxplot( )* is used to draw the boxplot. *face_wrap( )* makes a long ribbon of panels (generated by any number of variables) and wraps it into 2d. This is useful if you have a single variable with many levels and want to arrange the plots in a more space efficient manner.

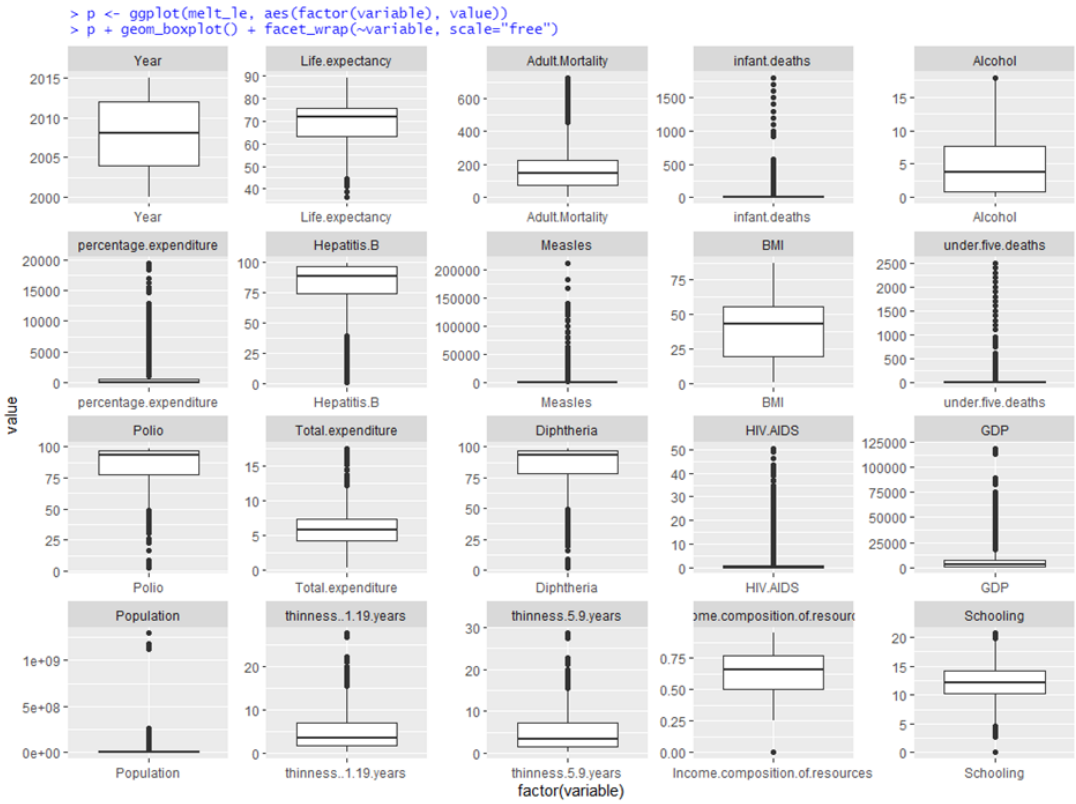

Figure 8: Boxplots with outliers

From the figure above, it shows that there are many outliers on almost all columns. The solution for this problem is by implementing winsorizing. winsorize( ) is a function from *DescTools* library which does the transformation of statistics by limiting extreme values in the statistical data to reduce the effect of possibly spurious outliers by assigning the outlier a lower weight or changing the value so that it is close to other values in the set. The basic method of understanding winsorizing:

1. Decide how much Winsorization you want. This is specified as a total percentage of untouched data. For example, if you want to Winsorize the top 5% and bottom 5% of data points, this is equal to 100% – 5% – 5% = 90% Winsorization. A 80% Winsorization means that 10% is modified from each tail area.
2. Replace the extreme values by the maximum and/or minimum values at the threshold. For example:
    - The following data set has several (bolded) extremes:

      {0.1, 1, 12, 14, 16, 18, 19, 21, 24, 26, 29, 32, 33, 35, 39, 40, 41, 44, 99, 125}

      Mean = 33.405.
    - After modifying the top and bottom 10% (I matched those values to the nearest extreme):

      {12, 12,12, 14, 16, 18, 19, 21, 24, 26, 29, 32, 33, 35, 39, 40, 41, 44, 44, 44}

      80% Winsorized mean = 24.95.

```
> # 3.3 Winsorize life expectancy
> le_cleanedv3$Life.expectancy <- Winsorize(le_cleanedv2$Life.expectancy,
+                                           minval = NULL, maxval = NULL,
+                                           probs = c(0.01, 0.99), na.rm = FALSE,
+                                           type = 7)
> # 3.4 Winsorize adult mortality
> le_cleanedv3$Adult.Mortality <- Winsorize(le_cleanedv2$Adult.Mortality,
+                                           minval = NULL, maxval = NULL,
+                                           probs = c(0.03, 0.97), na.rm = FALSE,
+                                           type = 7)
> # 3.5 Winsorize infant deaths
> le_cleanedv3$infant.deaths <- Winsorize(le_cleanedv2$infant.deaths,
+                                           minval = NULL, maxval = NULL,
+                                           probs = c(0.11, 0.89), na.rm = FALSE,
+                                           type = 7)
> # 3.6 Winsorize alcohol
> le_cleanedv3$Alcohol <- Winsorize(le_cleanedv2$Alcohol,
+                                   minval = NULL, maxval = NULL,
+                                   probs = c(0.001, 0.999), na.rm = FALSE,
+                                   type = 7)
> # 3.7 Winsorize percentage expenditure
> le_cleanedv3$percentage.expenditure <- Winsorize(le_cleanedv2$percentage.expenditure,
+                                   minval = NULL, maxval = NULL,
+                                   probs = c(0.14, 0.86), na.rm = FALSE,
+                                   type = 7)
> # 3.8 Winsorize Hepatitis B
> le_cleanedv3$Hepatitis.B <- Winsorize(le_cleanedv2$Hepatitis.B,
+                                           minval = NULL, maxval = NULL,
+                                           probs = c(0.09, 0.91), na.rm = FALSE,
+                                           type = 7)
> # 3.9 Winsorize measles
> le_cleanedv3$Measles <- Winsorize(le_cleanedv2$Measles,
+                                   minval = NULL, maxval = NULL,
+                                   probs = c(0.19, 0.81), na.rm = FALSE,
+                                   type = 7)
> # 3.11 Winsorize under five deaths
> le_cleanedv3$under.five.deaths <- Winsorize(le_cleanedv2$under.five.deaths,
+                           minval = NULL, maxval = NULL,
+                           probs = c(0.14, 0.86), na.rm = FALSE,
+                           type = 7)
> # 3.12 Winsorize polio
> le_cleanedv3$Polio <- Winsorize(le_cleanedv2$Polio,
+                                   minval = NULL, maxval = NULL,
+                                   probs = c(0.1, 0.9), na.rm = FALSE,
+                                   type = 7)
> # 3.13 Winsorize total expenditure
> le_cleanedv3$Total.expenditure <- Winsorize(le_cleanedv2$Total.expenditure,
+                           minval = NULL, maxval = NULL,
+                           probs = c(0.02, 0.98), na.rm = FALSE,
+                           type = 7)
> # 3.14 Winsorize Diphtheria
> le_cleanedv3$Diphtheria <- Winsorize(le_cleanedv2$Diphtheria,
+                                   minval = NULL, maxval = NULL,
+                                   probs = c(0.11, 0.89), na.rm = FALSE,
+                                   type = 7)
> # 3.15 Winsorize HIV/AIDS
> le_cleanedv3$HIV.AIDS <- Winsorize(le_cleanedv2$HIV.AIDS,
+                           minval = NULL, maxval = NULL,
+                           probs = c(0.19, 0.81), na.rm = FALSE,
+                           type = 7)
> # 3.16 Winsorize GDP
> le_cleanedv3$GDP <- Winsorize(le_cleanedv2$GDP,
+                           minval = NULL, maxval = NULL,
+                           probs = c(0.11, 0.89), na.rm = FALSE,
+                           type = 7)
> # 3.17 Winsorize Population
> le_cleanedv3$Population <- Winsorize(le_cleanedv2$Population,
+                           minval = NULL, maxval = NULL,
+                           probs = c(0.07, 0.93), na.rm = FALSE,
+                           type = 7)
> # 3.18 Winsorize thinness..1.19.years
> le_cleanedv3$thinness..1.19.years <- Winsorize(le_cleanedv2$thinness..1.19.years,
+                           minval = NULL, maxval = NULL,
+                           probs = c(0.04, 0.96), na.rm = FALSE,
+                           type = 7)
> # 3.19 Winsorize thinness.5.9.years
> le_cleanedv3$thinness.5.9.years <- Winsorize(le_cleanedv2$thinness.5.9.years,
+                           minval = NULL, maxval = NULL,
+                           probs = c(0.04, 0.96), na.rm = FALSE,
+                           type = 7)
```

```
> # 3.20 Winsorize Income.composition.of.resources
> le_cleanedv3$Income.composition.of.resources <- Winsorize(le_cleanedv2$Income.composition.of.resources,
+                           minval = NULL, maxval = NULL,
+                           probs = c(0.05, 0.95), na.rm = FALSE,
+                           type = 7)
> # 3.21 Winsorize Schooling
> le_cleanedv3$Schooling <- Winsorize(le_cleanedv2$Schooling,
+                           minval = NULL, maxval = NULL,
+                           probs = c(0.03, 0.97), na.rm = FALSE,
+                           type = 7)
```

Figure 9: Winsorize on columns

After that, the boxplot is plotted again to double check. Figure below shows there are no outliers left in the dataset.

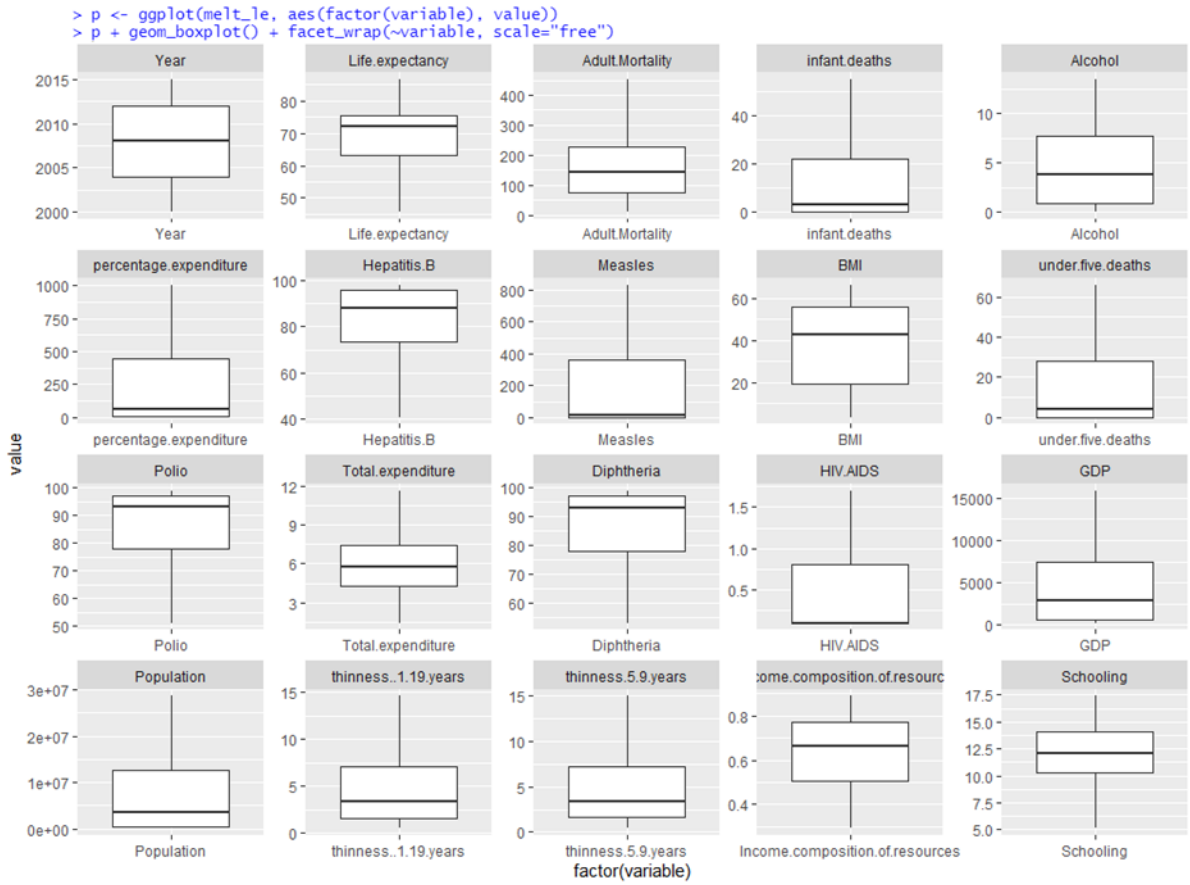

Figure 9: Boxplots without outliers

5. Data Visualization

5.1 Barplot

A bar plot is a graph that represents the category of data with rectangular bars with lengths and heights that is proportional to the values which they represent. It describes the comparisons between the discrete categories. One of the axis of the plot represents the specific categories being compared, while the other axis represents the measured values corresponding to those categories. For the x-axis, it will be Status and for the y-axis, it will

be Average Life Expectancy. From the figure below, the code uses *summarise( )* on grouped variable to create a new column while *ggplot( )* and *geom_bar( )* are used plot the barplot.

Figure 10: Barplot Status vs Avg Life Expectancy

5.2 Scatterplot

For the scatterplot, we will be plotting Life Expectancy on the y-axis and every other numeric column on the x-axis. *ggplot( )* and *geom_point( )* are used for plotting scatterplot while *geom_smooth( )* is used to draw a line to show linear relationship. Since there are 20 numeric columns, this report will be only showing three scatterplots with different type slopes. The scatterplot for the other graphs are available in the R file. From the Figure 11, it shows that there is a trend of positive slope between Income Composition of Resources and Life Expectancy. On Figure 12, it shows that the slope is undefined between Polio and Life Expectancy. Figure 13 shows that there is negative slope between Adult Mortality and Life Expectancy.

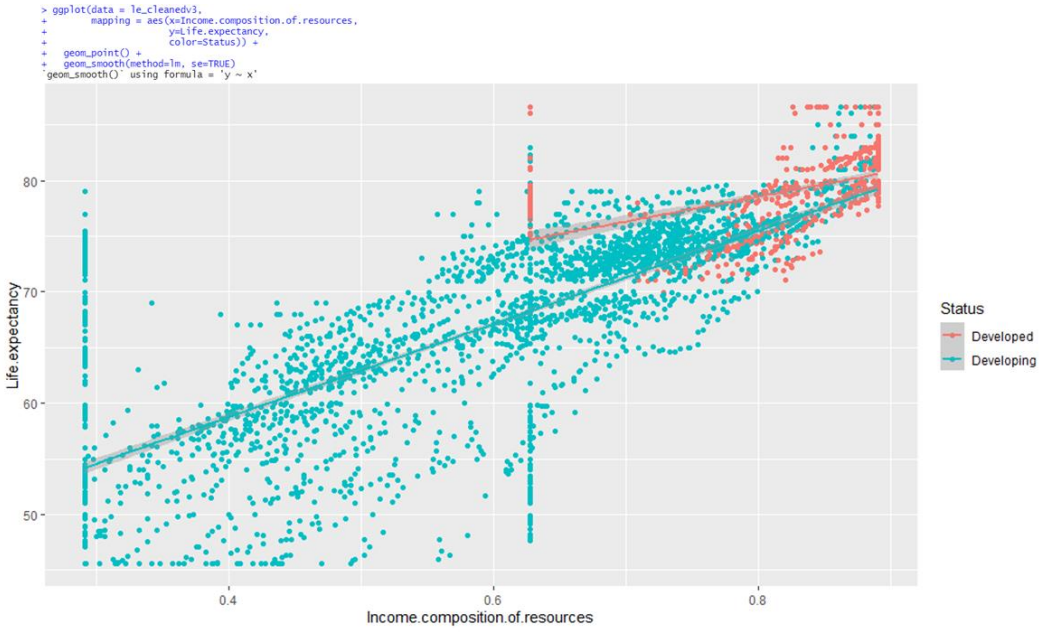

Figure 11: Scatterplot Income Composition of Resources vs Life Expectancy

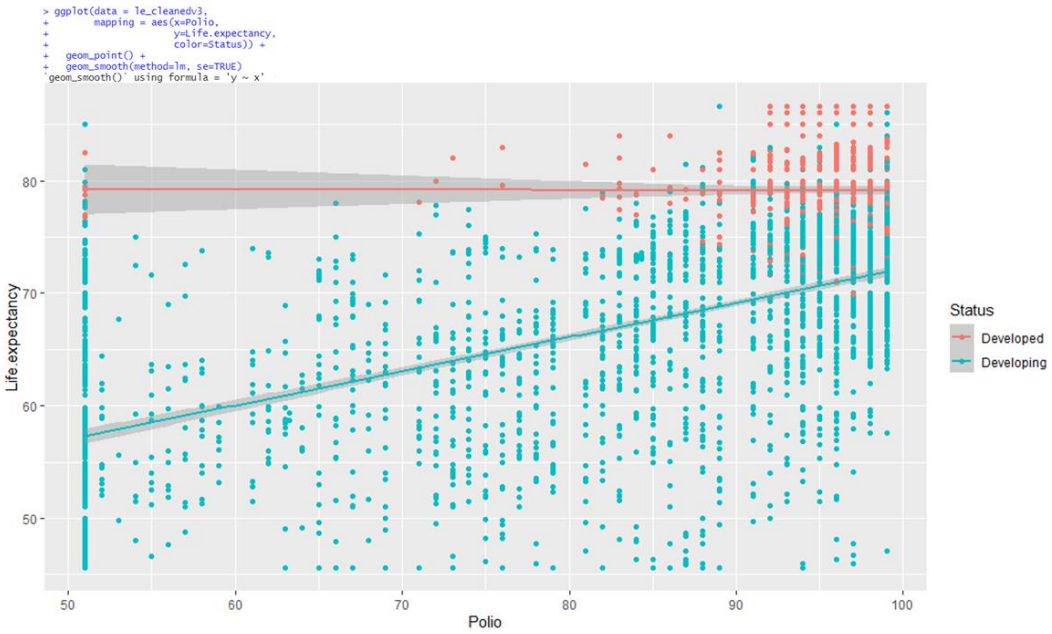

Figure 12: Scatterplot Polio vs Life Expectancy

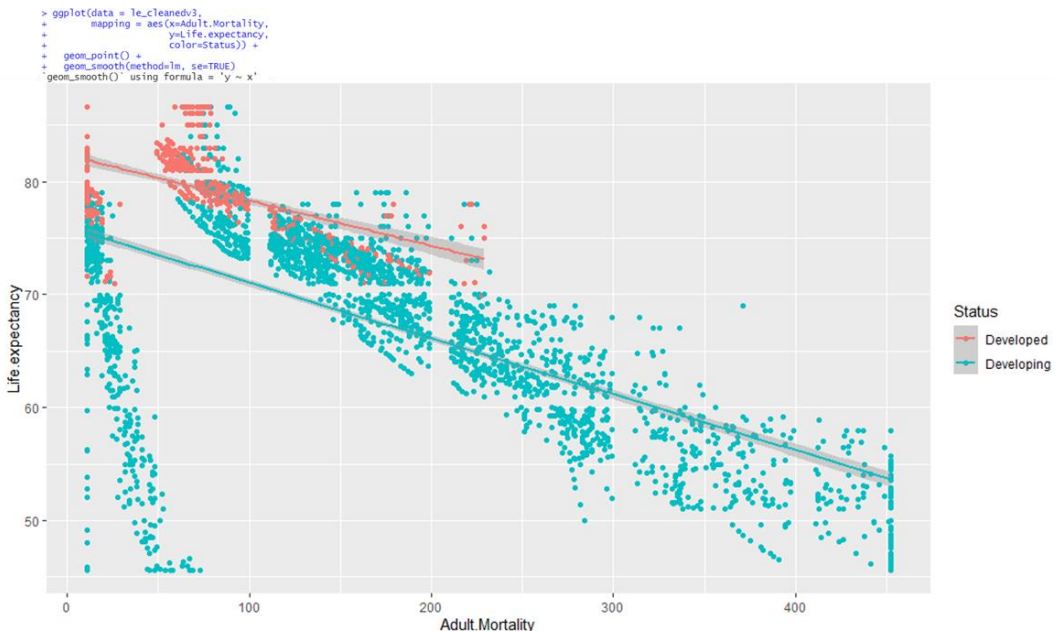

Figure 12: Adult Mortality vs Life Expectancy

6. Data Analysis

6.1 Correlation

From the objective of this project, data analysis will be doing on two columns, Income Composition of Resources as independent variable and Life Expectancy as dependent variable. The correlation between these two variables can be calculated using *cor.test( )* function. The output from Figure 13 shows the correlation between two variables is 0.7908784 which means it is a strong relationship between them since it is near to 1 and it is in positive direction.

```
> cor.test(le_cleanedv3$Life.expectancy, le_cleanedv3$Income.composition.of.resources)

        Pearson's product-moment correlation

data:  le_cleanedv3$Life.expectancy and le_cleanedv3$Income.composition.of.resources
t = 70.025, df = 2936, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.7769365 0.8040450
sample estimates:
      cor
0.7908784
```

Figure 13: Correlation code

6.2 Single Linear Regression

Because of only one x-axis and the relationship with y-axis is near to 1, the model will be created using is single linear regression. The function that will be using to create single linear regression model is *lm( ).* After that, *plot( )* and *abline( )* will be used to plot the data and draw straight best fit line. From Figure 14, the estimated intercept is 41.0992 and estimated slope for Income Composition of Resources is 43.9822. Figure 15 shows the graph of scatterplot and best fit line.

```
> slr <- lm(Life.expectancy ~ Income.composition.of.resources, data=datav3)
> summary(slr)

Call:
lm(formula = Life.expectancy ~ Income.composition.of.resources,
    data = datav3)

Residuals:
     Min       1Q   Median       3Q      Max
-21.0003  -2.0639   0.1824   2.5194  25.1020

Coefficients:
                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                      41.0992     0.4156   98.90   <2e-16 ***
Income.composition.of.resources  43.9822     0.6281   70.03   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.774 on 2936 degrees of freedom
Multiple R-squared:  0.6255,    Adjusted R-squared:  0.6254
F-statistic:  4904 on 1 and 2936 DF,  p-value: < 2.2e-16
```

Figure 14: Single linear regression model code

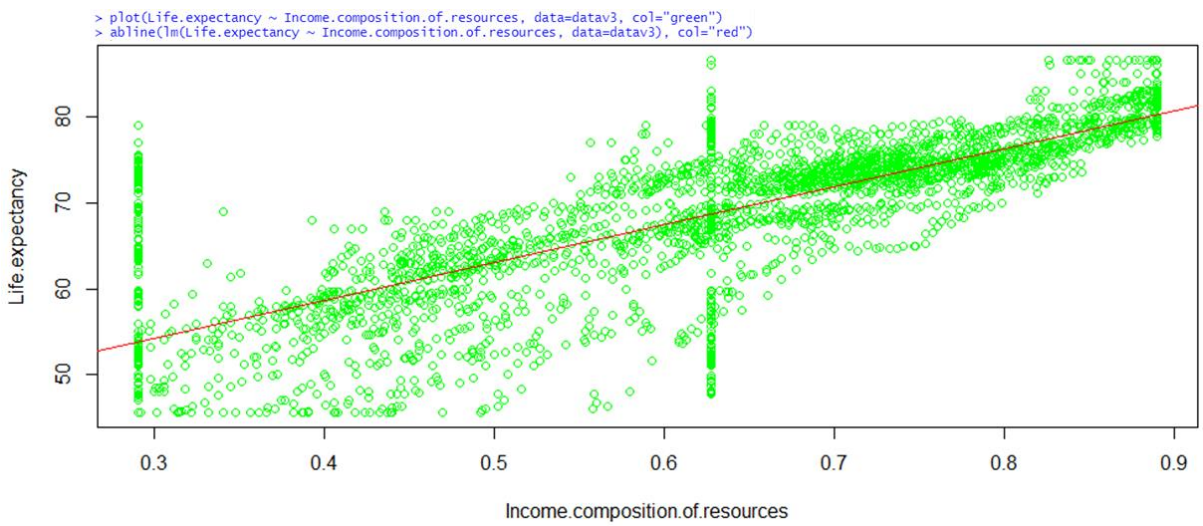

Figure 15: Single linear regression best fit line

6.3 Prediction

Next, the model will be using to do prediction to compare between the predicted y variable and actual y variable. From the figure below, a dataset is created by selecting 10 random rows from the cleaned dataset for only two columns, Income Composition of Resources and Life Expectancy. After that, the columns are renamed into 'X' and 'y'. Then, the new column 'y_' is created which is to store the predicted y value from the model using *predict( )* function. Lastly, residuals column is created in calculate the difference between actual y-values and predicted y-values.

```
> datav6 <- data.frame(le_cleanedv3[sample(nrow(le_cleanedv3), 10),
+                                   c('Income.composition.of.resources',
+                                     'Life.expectancy')])
> colnames(datav6) <- c("X","y")
> datav6["y_"] <- data.frame(y_ = c(predict(slr,data.frame(Income.composition.of.resources=datav6$X))))
> datav6["residuals"] <- abs(datav6$y - datav6$y_)
> datav6 <- data.frame(le_cleanedv3[sample(nrow(le_cleanedv3), 10),
+                                   c('Income.composition.of.resources',
+                                     'Life.expectancy')])
> colnames(datav6) <- c("X","y")
> datav6["y_"]<-data.frame(y_=c(predict(slr,
+                                       data.frame(Income.composition.of.resources=datav6$X))))
> datav6["residuals"] <- abs(datav6$y - datav6$y_)
> datav6
         X    y       y_  residuals
667  0.732 78.0 73.29417  4.7058317
208  0.459 65.3 61.28702  4.0129783
923  0.873 79.3 79.49566  0.1956616
2255 0.449 64.3 60.84720  3.4528005
2686 0.737 74.5 73.51408  0.9859206
2676 0.683 74.2 71.13904  3.0609606
2349 0.877 82.0 79.67159  2.3284096
1758 0.575 71.0 66.38896  4.6110406
796  0.693 74.4 71.57886  2.8211384
843  0.587 55.0 66.91675 11.9167461
```

Figure 16: Prediction code

7. Result

7.1 Formula Equation

The estimated intercept is 41.0992 and estimated slope for Income Composition of Resources is 43.9822. Therefore, the linear equation would be,

$$y = 41.0992 + 43.9822X$$

7.2 Conclusion

The conclusion that can be made from this project is, **"life expectancy expects to increase by one unit when income composition of resources increases by 43.9822 unit"**.

7.3 Lesson Learned and Improvement

The lesson learned throughout the project is applying steps on doing data analytics into real data world and self-study on exploring how to use functions to make code simpler but get more job done. The improvement can be made on this project is data cleaning on validating whether the value is realistic or not in the real world. This can help to get more accuracy on best fit line.